

Linxia GONG

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Eligible to work in Switzerland with permit B. No work permit sponsorship required from employers.

EDUCATION

Télécom ParisTech

Master's degree (Diplôme d'Ingénieur) in Computer Science, GPA 3.83/4.0

Nice, France
September 2015 – September 2017

Zhejiang University

Bachelor's Degree in Control Science and Engineering, GPA 3.92/4.0

Hangzhou, China
August 2011 – June 2015

EXPERIENCE

Sony AI Lab, Sony Europe

AI Engineer

November 2023 – Present
Zurich, Switzerland

- **Ethical Dataset Curation** | *Computer Vision, VLM* | Nature
 - * **Fairness Benchmark Dataset**: curated privacy-preserving, ethically licensed image dataset and evaluated bias in computer vision classification models and VLMs using this benchmark. Paper accepted at Nature.

Sea AI Lab, Sea Limited (Shopee's parent company)

Research Engineer

October 2021 – October 2023
Singapore

- **Multimodal Generative AI** | *LLM, 3D Reconstruction, PyTorch*
 - * **Language Models**: collected high-quality multilingual data to train LLM for Southeast Asian languages.
 - * **Speech-driven 3D Talking Avatar Synthesis**: reconstructed 3D representation of speaker from source video and animated a talking avatar based on input speech using NeRF and Diffusion models.
- **Reinforcement Learning and Game AI** | *Jax, Google Cloud Platform*
 - * **AlphaStar AI**: trained the AlphaStar AI model to achieve an 80% win rate against StarCraft II's built-in "Very Hard" level AI.
 - * **RL Engineering Support**: worked on module development and framework refactoring, for applying state-of-art techniques and improving time and memory efficiency; supported experiment deployment on TPUs and GPUs.

Fuxi AI Lab, NetEase Games (2nd biggest game company in China)

Data Mining Research Engineer

December 2017 – October 2021
Hangzhou, China

- **Anti-Fraud Detection** | *Sequence Embedding, Graph Embedding, Keras* | KDD
 - * **Anomalous Behavior Detection**: detected cheating players and provide evidence to justify punishments based on sequential behaviors. Paper accepted at KDD'2018.
 - * **Other Detection**: supported the development of risky trade detection, trajectory detection and hack detection.
- **Players Matchmaking** | *Graph Embedding, Explainable AI, DQN, Operation Research* | KDD, GDC, CIKM
 - * **Matchmaking Services**: designed and deployed matchmaking systems that arrange players into teams to maximize their overall engagement in player-versus-player competitions. Services deployed to 13 gameplays with an average 10% reduction in churn rate. Papers accepted at KDD'2020 and KDD'2021; work presented at GDC'2021.
 - * **Win and User Engagement Prediction Models**: built models to estimate the matchmaking quality, based on graph embedding algorithms and game theory methods like SHAP. Several new win/churn patterns were discovered.

PUBLICATIONS

Nature	Fair human-centric image dataset for ethical AI benchmarking
KDD'2021	Globally Optimized Matchmaking in Online Games
KDD'2020 (Oral)	OptMatch: Optimized Matchmaking via Modeling the High-Order Interactions on the Arena
CIKM'2020	Match Tracing: A Unified Framework for Real-time Win Prediction and Quantifiable Performance
CIKM'2019	GMTL: A GART Based Multi-task Learning Model for Multi-Social-Temporal Prediction in Online Games
KDD'2018	NGUARD: A Game Bot Detection Framework for NetEase MMORPGs

TECHNICAL SKILLS

Programming Languages: Python+cuda, Cython, Shell (Linux), SQL, HTML/CSS+JavaScript
Frameworks and Libraries: PyTorch, Jax, Keras, TensorFlow, NumPy, Matplotlib, OpenCV, FFmpeg
Developer Tools: Git, Kubernetes/Ray/Docker, AWS, GCP (Google Cloud Platform), CI/CD, Apache Flink
Languages: Chinese (native), English (fluent), French (intermediate)